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Time Keeping Errors



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December 31, 2008

Zune HD Portable Media Player

Leap year was not handled correctly by the internal clock driver.

Devices froze for 24 hours.

Solution was to completely drain the battery and then recharge after 12:00pm on January 1, 2009.



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February, 2007

F-22A Raptor

A squadron of six U.S. Air Force F-22 Raptor stealth fighters embarked on a 12 to 15-hour flight from Hawaii to Okinawa.

They crossed the international date line.

Maj. Gen. Don Sheppart (ret.) "... At the international data line, whoops, all systems dumped and when I say all systems, I mean all systems, their navigation, part of their communications, their fuel systems. They were – they could have been in real trouble. They were with their tankers. The tankers – they tried to reset their systems, couldn't get them reset. The tankers brought them back to Hawaii. This could have been real serious.It certainly could have been real serious if the weather had been bad. It turned out OK. It was fixed in 48 hours."



Fair use, https://defenseindustrydaily.com/wp-content/uploads/2023/11/AIR_F-22_Fuel_Tanks_Landing_lg.jpg

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February 25, 1991

Patriot Air Defense System

Internal clock drift by 0.33 second over 100 hours of operation
– equivalent to 600 meters of target drift at target velocities

Radar detected the Scud missile and predicted where to find it next

Radar pulse times were converted to floating point differently
– one was correct, the other off by 0.33 seconds or 600m

Target was marked as spurious and removed from tracking

Iraqi Al Hussein Scud missile hit barracks in Dhahran, Saudi Arabia

28 U.S. soldiers were killed



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